



Python Data Science Cheat Sheet

1. What Is Data Science?

Data science is about using **data** to find patterns, make predictions, and solve problems. It combines **math**, **coding**, and **creativity** to understand the world better.

2. Getting Started

Install Python Libraries

```
pip install numpy pandas matplotlib seaborn scikit-learn
```

Import Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

3. Working with Data

Reading Data

```
df = pd.read_csv('data.csv') # Load a CSV file
df.head()                   # Show first 5 rows
df.info()                   # Show data info
df.describe()               # Show summary statistics
```

Selecting Data

```
df['column']                # Select one column
df[['col1', 'col2']]        # Select multiple columns
df.iloc[0]                  # Select first row
df.loc[0, 'col1']           # Select specific cell
```

Filtering Data

```
df[df['age'] > 15]          # Students older than 15  
df[(df['grade'] > 80) & (df['gender'] == 'F')]
```

4. Cleaning Data

```
df.isnull().sum()          # Check missing values  
df.dropna(inplace=True)    # Remove missing rows  
df.fillna(0, inplace=True) # Replace missing with 0  
df.rename(columns={'old': 'new'}, inplace=True)
```

5. Numpy Basics

```
arr = np.array([1, 2, 3, 4])  
arr.mean()                 # Average  
arr.std()                  # Standard deviation  
np.arange(0, 10, 2)        # Evenly spaced numbers  
np.linspace(0, 1, 5)       # 5 numbers between 0 and 1
```

6. Visualizing Data

Matplotlib

```
plt.plot(df['x'], df['y'])  
plt.title('Line Chart')  
plt.xlabel('X-axis')  
plt.ylabel('Y-axis')  
plt.show()
```

Seaborn

```
sns.histplot(df['age'], bins=10, color='skyblue')  
sns.boxplot(x='gender', y='score', data=df)  
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
```

7. Understanding Data

Grouping and Sorting

```
df.groupby('gender')['score'].mean() # Average score by gender
df.sort_values('score', ascending=False)
```

Creating New Columns

```
df['total'] = df['math'] + df['science'] + df['english']
df['average'] = df['total'] / 3
```

8. Machine Learning Basics

Splitting Data

```
from sklearn.model_selection import train_test_split
X = df[['feature1', 'feature2']]
y = df['target']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

Training a Model

```
from sklearn.linear_model import LinearRegression
model = LinearRegression()
model.fit(X_train, y_train)
predictions = model.predict(X_test)
```

Evaluating the Model

```
from sklearn.metrics import mean_squared_error, r2_score
mse = mean_squared_error(y_test, predictions)
r2 = r2_score(y_test, predictions)
print(mse, r2)
```

9. Common Data Science Libraries

Library	Purpose
NumPy	Math and arrays
Pandas	Data tables and cleaning
Matplotlib	Basic charts
Seaborn	Stylish charts
Scikit-learn	Machine learning
Statsmodels	Statistics and modeling

10. Tips for Students

- Start small — use simple datasets like grades or sports scores.
- Always clean your data before analyzing it.
- Use graphs to make your results easy to understand.
- Experiment with colors and chart types to make visuals fun.
- Practice regularly — coding gets easier with time.

Key Idea:

Data science is like solving puzzles with numbers. Python helps turn data into stories that explain how things work and why they happen.